ENVIRONMENTAL SOUND CLASSIFICATION / REPORT 01

Environmental sound classification

Complete Colab matrix results

The latest executed study: architectures, training augmentation, representations, sampling rates and synthetic stress. Every score in the main results is development validation. Historical test exposure prevents a new independent-confirmation claim.

65 / 65	60 × 60 epochs	7
Audited tasks	Neural training runs	Embedded figures

Prepared 29 October 2026. Main recorded experiments: 28 October 2026.

All figures are embedded at their original resolution in the full-page figure atlas. Click any figure reference to jump to its page. Full tables and scientific limitations are retained. Companion-project evidence paths are indexed at the end.

This report documents internal agent-produced execution and verification. It does not certify external human replication, learner mastery, state of the art or production readiness.

Finalized: 29 October 2026; main experiments ran on 28 October. This report covers the supplied Downloads project and standalone Study Matrix. **65/65 tasks were executed, downloaded and audited**, including 60 neural runs completing 60 epochs each. This is development-validation research; historical test exposure prevents a claim of fresh independent confirmation.

The architecture group with the highest validation mean is **BiGRU + Channel Attention: 74.37%**, +0.50 percentage points relative to the original BiGRU. The highest feature-track mean is `sr32_fmax16_logmel32` at 57.12%; that track uses a different pooled CNN and is not a single-variable replacement for the main architecture. At 0 dB Gaussian-noise SNR, noise-trained versus clean-trained CNN changes accuracy by **+19.35 points**. All conditions are reported below.

Detailed English content guide [P01]

1. Review findings and repairs

Severity	Demonstrated issue	Verified repair
P1	Stale directories could impersonate completed tasks	Check task/config/cache/source/checkpoint identity, full history and prediction metrics
P1	ID-only matrix checks ignored semantic changes	Consume matrix with schema, split, budget and semantic validation
P1	Silent subprocesses could evade timeout	Poll output/process and terminate its group at command/session deadline
P1	Feature aggregation admitted incomplete/fabricated records	Validate full exposure, selected epoch, weights, membership/labels and metrics; publish completion last
P1	Live Colab kernel did not discover editable installation	Add verified source directory to live kernel; pilot rerun succeeded
P2	Results view omitted stress.json	Add explicit three-model × six-condition result table
P1	CI omitted baseline dependencies required by its tests	Install dev, baseline and report extras in CI and current guide

The repaired project passed 274 tests locally and 274 in Colab; the final notebook contains no error outputs. Initial issues were a missing local scikit-learn dependency, omitted historical test fixtures in our first cloud upload, and a live-kernel import error after CNN pilot training. Logs are retained; failed execution was not counted as success. CI configuration was repaired, but no GitHub Actions run is claimed.

2. Executed scope and data protocol

Group	Tasks	Executed content
A	2	Log-Mel statistics + logistic; cepstral statistics + LinearSVC
B	15	CNN, BiGRU, TA, CA, TCA × three seeds
C	9	SpecAugment, waveform noise, waveform Mixup × three seeds
D	36	12 feature/sampling cases × three seeds
E	3	Six validation stress conditions for each of three seed101 checkpoints

Three two-epoch pilots and one real-data resume probe are stored separately and excluded from the 60 main runs and their means. PANNs/AST/BEATs/CLAP/RL are not implemented; no execution or score is invented. No new fold 5 evaluation or confirmation CV was performed. Software tests check old fixtures, which is distinct from evaluating new models on test audio.

The pinned ESC-50 data contains 2,000 five-second clips in 50 classes. Original WAVs were downloaded and a v2 cache prepared; all 2,000 cached identities/labels align with original metadata. Train folds 1-3, validation fold 4 and designated test role fold 5 become **1200 / 398 / 400** after source-strict exclusions. Validation classes 6 and 37 have seven clips each; all others have eight. Normalization is fitted only on training clips. Exact cross-fold duplicate checks passed; perceptual near-duplicate auditing is incomplete.

Main input: 16 kHz mono → 512-sample Hann STFT, hop 256 → 64 HTK Mel-power bands → natural log → train-only band normalization, shape `[B,1,64,313]`. The encoder outputs `[B,64,8,39]`; GRU input is `[B,39,512]`, with 64 hidden units per direction and 128 output features. TA learns softmax time weights; CA uses 64 → 8 → 64 gates with 2 × sigmoid and identity initialization.

Neural runs use 60 epochs, batch size 32, AdamW (learning rate 0.001, weight decay 0.0001), cosine scheduling and gradient clipping at 5. Main training records 1,200 examples and 38 updates per epoch, or 2,280 updates in total. Every run receives the same 60-epoch budget; selected weights may come from an earlier epoch with the earliest minimum validation cross-entropy. Logistic also selects by validation cross-entropy; SVM selects accuracy, then macro-F1 and smaller C. Accuracy and equally weighted 50-class macro-F1 are recalculated from per-clip predictions.

3. Classical baselines and architecture results

Baseline	Accuracy	Macro-F1	Selected parameter
logistic	42.21%	38.86%	0.001
cepstral_svm	40.20%	37.50%	1.0

Logistic uses 128 temporal Log-Mel means/standard deviations; SVM uses 40 statistics from 20 DCT coefficients. Features, classifier and selection rule differ, so the score difference cannot be attributed to classifier choice alone. Candidate convergence was checked.

Model	Accuracy: 101 / 202 / 303	Accuracy mean ± SD (%)	Macro-F1 mean ± SD (%)	Parameters
CNN	72.61%, 72.36%, 72.86%	72.61 ± 0.25	71.39 ± 0.58	49058
CNN + BiGRU	75.13%, 73.87%, 72.61%	73.87 ± 1.26	72.97 ± 1.36	251810
BiGRU + Temporal Attention	74.87%, 70.10%, 71.86%	72.28 ± 2.41	71.52 ± 2.41	251939
BiGRU + Channel Attention	75.38%, 74.62%, 73.12%	74.37 ± 1.15	73.48 ± 1.00	252906
BiGRU + Temporal + Channel Attention	74.12%, 71.61%, 72.86%	72.86 ± 1.26	71.69 ± 1.35	253035

Figure 1. Three-seed development validation; SD is not a confidence interval. - see the full-page figure atlas.

Three-seed development validation; SD is not a confidence interval. Data [P02] · Plot code [P03]

All six new CNN/GRU runs match the historical development selected epochs, every validation prediction and weight tensors exactly. This is regression evidence from actual reruns, not reused old scores relabeled as new experiments.

4. Augmentation and paired changes

Training condition	Accuracy mean ± SD (%)	Macro-F1 mean ± SD (%)
Clean CNN	72.61 ± 0.25	71.39 ± 0.58
specaugment	71.86 ± 1.15	70.75 ± 1.38
noise	73.12 ± 0.87	71.99 ± 1.23
mixup	71.61 ± 0.66	70.65 ± 0.44

Figure 2. Augmentation affects training only; validation recordings stay clean. - see the full-page figure atlas.

Augmentation affects training only; validation recordings stay clean. Data [P02] · Plot code [P03]

SpecAugment masks normalized time/frequency regions; noise adds Gaussian corruption with probability 0.5 at 10-30 dB; Mixup combines waveforms and both label losses with Beta(0.4, 0.4). Augmented and clean training losses are different objectives and are not directly ranked against one another.

Paired contrast	Per-seed accuracy delta (pp)	Mean ± SD (pp)	Mean macro-F1 delta (pp)
attention_ca_vs_gru	+0.25, +0.75, +0.50	+0.50 ± 0.25	+0.51
attention_interaction	-1.01, +0.75, +0.50	+0.08 ± 0.95	-0.33
attention_ta_vs_gru	-0.25, -3.77, -0.75	-1.59 ± 1.90	-1.45
attention_tca_vs_gru	-1.01, -2.26, +0.25	-1.01 ± 1.26	-1.28
gru_vs_cnn	+2.51, +1.51, -0.25	+1.26 ± 1.40	+1.58
mixup_vs_clean_cnn	-0.50, -1.51, -1.01	-1.01 ± 0.50	-0.74
noise_vs_clean_cnn	+0.00, +1.76, -0.25	+0.50 ± 1.10	+0.60
specaugment_vs_clean_cnn	+0.25, -1.76, -0.75	-0.75 ± 1.01	-0.64

Figure 3. Within-seed deltas; these are not significance tests. - see the full-page figure atlas.

Within-seed deltas; these are not significance tests. Data [P04] · Plot code [P03]

Attention interaction is TCA - TA - CA + original GRU, describing deviation from additive module effects rather than proving an unconfounded causal mechanism. Three seeds reuse the same 398 validation clips; they are not independent datasets, and seed SD is not a dataset-sampling confidence interval.

5. Twelve feature and sampling-rate conditions

Case	Accuracy mean ± SD (%)	Macro-F1 mean ± SD (%)
sr16_fmax8_loglinear64	51.59 ± 0.81	48.58 ± 1.15
sr16_fmax8_logmel32	55.61 ± 0.73	52.97 ± 0.48
sr16_fmax8_logmel64	51.59 ± 2.05	48.46 ± 2.20
sr16_fmax8_mfcc40	49.25 ± 1.74	47.08 ± 1.75

Case	Accuracy mean ± SD (%)	Macro-F1 mean ± SD (%)
sr32_fmax16_loglinear64	48.99 ± 1.33	46.09 ± 1.74
sr32_fmax16_logmel32	57.12 ± 1.05	54.59 ± 1.52
sr32_fmax16_logmel64	51.09 ± 0.38	48.33 ± 0.48
sr32_fmax16_mfcc40	49.58 ± 3.10	47.29 ± 2.93
sr32_fmax8_loglinear64	50.75 ± 0.66	47.58 ± 1.57
sr32_fmax8_logmel32	55.70 ± 1.05	53.00 ± 1.34
sr32_fmax8_logmel64	51.26 ± 1.96	48.23 ± 2.16
sr32_fmax8_mfcc40	49.83 ± 1.71	47.63 ± 1.73

Figure 4. Track D uses the 26,658-parameter pooled CNN; interpret separately from the main architecture track. - see the full-page figure atlas.

Track D uses the 26,658-parameter pooled CNN; interpret separately from the main architecture track. Data [P02] · Plot code [P03]

Representations are Log-Mel 64, Log-Mel 32, the first 40 orthonormal DCT-II coefficients of 64-band Log-Mel, and Log-linear 64. Conditions are 16k/F8, 32k/F8 and 32k/F16, with 32 ms windows, 16 ms hops and 313 frames. The 16k/F8 → 32k/F8 comparison matches usable bandwidth; 32k/F8 → 32k/F16 adds high frequencies and changes filter spacing. MFCC changes row count and coordinate meaning. Equal parameter count does not imply equal computation or a pure single-variable causal experiment. All 17 × 3 within-track paired contrasts are retained in CSV; missing scores were never imputed as zero.

6. Noise and gain sensitivity

Condition	Clean-trained CNN	Clean-trained BiGRU	Noise-trained CNN
clean	72.61%	75.13%	72.61%
snr_20	47.74%	54.77%	70.60%
snr_10	23.37%	24.62%	62.81%
snr_0	6.78%	6.28%	26.13%
gain_-6	68.84%	69.85%	68.59%
gain_6	71.86%	73.37%	70.60%

Figure 5. 398 clips per cell; one selected seed101 checkpoint per model. - see the full-page figure atlas.

398 clips per cell; one selected seed101 checkpoint per model. Data [P05] · Plot code [P03]

These are deterministic, per-clip RMS-scaled Gaussian-noise and gain interventions, with identical realizations across models and no peak clipping. 0 dB equates added-noise and original-recording RMS power; the original ambient SNR is unknown. Improvements on these conditions do not establish robustness to field noise, rooms, devices or unknown classes. Track E covers three checkpoints, not three-seed robustness.

7. Integrity, replay and efficiency evidence

An independent script recalculates 50-class confusion matrices, accuracy and macro-F1 from CSVs, checking all 65 tasks, full epoch budgets, selected epochs and configurations. Colab then re-executes the project frontends and models from original validation WAVs: **62 fitted models × 398 clips = 24,676 predictions all reproduce**. This is actual forward replay rather than reading saved scores. A real-data probe pauses after one epoch and resumes for epoch 2, matching uninterrupted pilot weights bit-for-bit. Resume evidence applies to the same runtime/code/data, not arbitrary versions or devices.

Model	CPU WAV-to-logits p95(ms)	T4 WAV-to-logits p95(ms)	Checkpoint(KiB)
CNN	25.787	16.045	281.1
CNN + BiGRU	24.477	17.860	1075.4
BiGRU + Temporal Attention	32.398	16.501	1076.5
BiGRU + Channel Attention	25.846	17.290	1080.9
BiGRU + Temporal + Channel Attention	35.366	17.276	1081.9

Measured on Colab `Intel(R) Xeon(R) CPU @ 2.00GHz` / `Tesla T4` with 2 CPU threads, batch size 1, float32, 10 warmups, 100 repetitions and one fixed validation WAV. CUDA is synchronized. WAV-to-logits includes decoding, frontend and transfer, excluding model loading, process startup and service queues. Torch peak allocated memory is not total driver memory. Recorded training-loop wall times for 60 runs sum to 72.75 minutes, including in-loop validation/checkpoint writes but excluding initialization, cache/WAV loading and verification, acquisition/preparation, pilots/probes, idle and export; this is not total session time. Paid spend was 0, with no upgrade or resource purchase.

Figure 6. Fixed-seed101 validation curves; checkpoints use minimum cross-entropy. - see the full-page figure atlas.

Fixed-seed101 validation curves; checkpoints use minimum cross-entropy. Data [P06] · Plot code [P03]

Per-class error slices: fixed seed101

Model	Zero-recall classes	Three lowest-recall classes:correct/support
CNN	0	sheep 2/8; wind 3/8; clapping 3/8
CNN + BiGRU	0	helicopter 1/8; door_wood_creaks 2/8; sheep 3/8
BiGRU + Temporal Attention	0	door_wood_creaks 1/8; helicopter 1/8; wind 3/8
BiGRU + Channel Attention	0	helicopter 1/8; drinking_sipping 2/8; door_wood_creaks 2/8
BiGRU + Temporal + Channel Attention	0	drinking_sipping 1/8; helicopter 1/8; wind 2/8

Each class has only 7 or 8 validation clips, so one error substantially changes per-class recall. These slices are exploratory diagnostics. Every run retains its full confusion matrix and per-clip predictions; an aggregate score must not conceal class-specific failures.

8. Limits and next decision

This round establishes architecture, augmentation, representation and synthetic-stress behavior within the declared development protocol, not SOTA, algorithmic novelty or production readiness. Attention is not guaranteed to help; assess all seeds and small effects instead of the highest single score. Equal epoch budgets do not match parameters, FLOPs and latency simultaneously. The bidirectional GRU is an offline model, not evidence of causal streaming recognition. Attention weights are not reliable causal explanations.

Do not subtract the old roughly 62% test result from the new 70%+ validation results and call the gap improvement. They measure different recordings and purposes. Historical fold 5 three-seed means are 63.08% for CNN and 63.92% for GRU, referenced only as old evidence; this round does not reconfirm them. All ESC-50 folds have prior exposure, so further selection remains exploratory. Confirmation requires a new source-isolated evaluation boundary and defined real-noise needs; renaming an old fold cannot restore blindness.

The old CNN bundle remains the historical default. All new models are delivered as research checkpoints, retaining negative findings and failure logs. CV exam learning records were untouched; agent-produced work is not presented as user mastery, external human replication or institutional certification.

9. Evidence and reproduction

Per-run results [P07] · Three-seed aggregates [P02] · 75 paired deltas [P04] · Stress results [P05] · Complete audit [P08] · WAV replay [P09] · Resume check [P10] · Timing definition [P11]

Environment: torch=2.11.0+cu130, numpy=2.1.3, scipy=1.16.3, soundfile=0.14.0, sklearn=1.6.1.

```
python -m pip install -e '.[dev,baseline,report]'
python -m esc50_sound prepare
python -m pytest -q
python tools/analyze_colab_matrix.py --require-complete
python tools/plot_colab_matrix.py
```

The full matrix uses `notebooks/ESC50_Matrix_Full_20261028.ipynb`; it trains all tasks, so read its budget settings first. The safe preview remains plan mode in `ESC50_Colab_Research_Console.ipynb`. Feature runs cannot resume by epoch; main training requires matching environment and last.pt. Colab disk is temporary; this round exported/downloaded every task. A saved notebook is not a copy of all weights or audio.

Data and license: official ESC-50, pinned commit `33c8ce9eb2cf0b1c2f8bcf322eb349b6be34dbb6`. Overall dataset license is CC BY-NC 3.0; retain credits. This project grants no commercial rights. See Colab resource information. Primary method references and reading depth appear in the literature reading notes [P12]; actual saved runs and per-clip outputs directly support this round's results.

Saved Colab reader and final review coverage

Open the bilingual results in Colab. Downloaded readback matches every source cell, stream output and all four plot images; six code cells contain no errors. The reader displays the saved audited study without retraining.

Review coverage and remaining limits [P13] · Measurement environment [P14]

Appendix A. Real audio and Log-Mel input

The example is the validation recording `4-102844-A-49.wav` (hand saw). It illustrates the waveform and the natural-log-power Mel representation used by the main models. Amplitude is a digital signal value, not calibrated sound-pressure level; the Mel axis is an index over nonuniform frequency bands.

Figure 7. A real validation waveform and Log-Mel representation - see the full-page figure atlas.

Appendix B. All fitted-task validation results

All 62 fitted tasks are listed: two classical models and 60 neural runs. Each neural run completed 60 epochs; the displayed checkpoint is selected by the earliest minimum validation cross-entropy. The three remaining E tasks evaluate already fitted checkpoints; their six conditions each are reported in Section 6.

Task ID	Accuracy (%)	Macro-F1 (%)	Selected epoch
A_logistic	42.21	38.86	not applicable
A_cepstral_svm	40.20	37.50	not applicable
B_cnn_freq_s101	72.61	71.94	47
B_cnn_freq_s202	72.36	70.79	49
B_cnn_freq_s303	72.86	71.45	46
B_crnn_freq_s101	75.13	74.21	58
B_crnn_freq_s202	73.87	73.18	58
B_crnn_freq_s303	72.61	71.52	30
B_crnn_freq_ta_s101	74.87	74.02	46
B_crnn_freq_ta_s202	70.10	69.22	58
B_crnn_freq_ta_s303	71.86	71.31	40
B_crnn_freq_ca_s101	75.38	74.38	45
B_crnn_freq_ca_s202	74.62	73.64	58
B_crnn_freq_ca_s303	73.12	72.41	57
B_crnn_freq_tca_s101	74.12	72.86	28
B_crnn_freq_tca_s202	71.61	70.21	58
B_crnn_freq_tca_s303	72.86	72.02	40
C_specaugment_s101	72.86	71.96	45
C_specaugment_s202	70.60	69.25	58
C_specaugment_s303	72.11	71.04	38
C_noise_s101	72.61	71.28	57
C_noise_s202	74.12	73.41	41
C_noise_s303	72.61	71.29	52
C_mixup_s101	72.11	71.16	58
C_mixup_s202	70.85	70.40	56
C_mixup_s303	71.86	70.41	57
D_sr16_fmax8_logmel64_s101	49.25	46.02	59

Task ID	Accuracy (%)	Macro-F1 (%)	Selected epoch
D_sr16_fmax8_logmel64_s202	52.51	49.08	49
D_sr16_fmax8_logmel64_s303	53.02	50.27	57
D_sr16_fmax8_logmel32_s101	54.77	52.42	58
D_sr16_fmax8_logmel32_s202	56.03	53.28	58
D_sr16_fmax8_logmel32_s303	56.03	53.21	57
D_sr16_fmax8_mfcc40_s101	48.24	45.71	58
D_sr16_fmax8_mfcc40_s202	51.26	49.05	56
D_sr16_fmax8_mfcc40_s303	48.24	46.48	59
D_sr16_fmax8_loglinear64_s101	51.01	47.65	58
D_sr16_fmax8_loglinear64_s202	51.26	48.22	49
D_sr16_fmax8_loglinear64_s303	52.51	49.87	57
D_sr32_fmax8_logmel64_s101	48.99	45.78	59
D_sr32_fmax8_logmel64_s202	52.26	49.10	49
D_sr32_fmax8_logmel64_s303	52.51	49.83	57
D_sr32_fmax8_logmel32_s101	56.53	54.17	58
D_sr32_fmax8_logmel32_s202	54.52	51.54	58
D_sr32_fmax8_logmel32_s303	56.03	53.31	59
D_sr32_fmax8_mfcc40_s101	48.49	46.16	58
D_sr32_fmax8_mfcc40_s202	51.76	49.53	56
D_sr32_fmax8_mfcc40_s303	49.25	47.18	59
D_sr32_fmax8_loglinear64_s101	50.25	46.63	58
D_sr32_fmax8_loglinear64_s202	50.50	46.72	49
D_sr32_fmax8_loglinear64_s303	51.51	49.39	57
D_sr32_fmax16_logmel64_s101	50.75	47.91	58
D_sr32_fmax16_logmel64_s202	51.51	48.84	49
D_sr32_fmax16_logmel64_s303	51.01	48.23	57
D_sr32_fmax16_logmel32_s101	56.78	54.24	58
D_sr32_fmax16_logmel32_s202	56.28	53.28	56
D_sr32_fmax16_logmel32_s303	58.29	56.26	57
D_sr32_fmax16_mfcc40_s101	46.98	44.76	54

Task ID	Accuracy (%)	Macro-F1 (%)	Selected epoch
D_sr32_fmax16_mfcc40_s202	53.02	50.50	56
D_sr32_fmax16_mfcc40_s303	48.74	46.62	59
D_sr32_fmax16_loglinear64_s101	47.99	45.38	58
D_sr32_fmax16_loglinear64_s202	48.49	44.81	49
D_sr32_fmax16_loglinear64_s303	50.50	48.07	57

Appendix C. All aggregated paired contrasts

Each row averages three within-seed contrasts on the same 398 clips. The delta is right minus left, except the separately defined attention interaction TCA - TA - CA + original GRU. SD is training-seed spread, not a dataset confidence interval.

Contrast (right minus left)	Accuracy delta: mean ± SD (pp)	Macro-F1 delta (pp)
added_high_frequency_bandwidth; D_sr32_fmax8_loglinear64 → D_sr32_fmax16_loglinear64	-1.759 ± 0.665	-1.495
added_high_frequency_bandwidth; D_sr32_fmax8_logmel32 → D_sr32_fmax16_logmel32	+1.424 ± 1.046	+1.590
added_high_frequency_bandwidth; D_sr32_fmax8_logmel64 → D_sr32_fmax16_logmel64	-0.168 ± 1.710	+0.090
added_high_frequency_bandwidth; D_sr32_fmax8_mfcc40 → D_sr32_fmax16_mfcc40	-0.251 ± 1.399	-0.334
attention_ca_vs_gru; B_crnn_freq → B_crnn_freq_ca	+0.503 ± 0.251	+0.508
attention_interaction; B_crnn_freq_ta+B_crnn_freq_ca-B_crnn_freq → B_crnn_freq_tca	+0.084 ± 0.951	-0.333
attention_ta_vs_gru; B_crnn_freq → B_crnn_freq_ta	-1.591 ± 1.902	-1.452
attention_tca_vs_gru; B_crnn_freq → B_crnn_freq_tca	-1.005 ± 1.256	-1.276
gru_vs_cnn; B_cnn_freq → B_crnn_freq	+1.256 ± 1.399	+1.577
mixup_vs_clean_cnn; B_cnn_freq → C_mixup	-1.005 ± 0.503	-0.739
noise_vs_clean_cnn; B_cnn_freq → C_noise	+0.503 ± 1.095	+0.599
representation_vs_logmel64; D_sr16_fmax8_logmel64 → D_sr16_fmax8_loglinear64	+0.000 ± 1.569	+0.121
representation_vs_logmel64; D_sr16_fmax8_logmel64 → D_sr16_fmax8_logmel32	+4.020 ± 1.330	+4.514
representation_vs_logmel64; D_sr16_fmax8_logmel64 → D_sr16_fmax8_mfcc40	-2.345 ± 2.107	-1.377
representation_vs_logmel64; D_sr32_fmax16_logmel64 → D_sr32_fmax16_loglinear64	-2.094 ± 1.384	-2.238

Contrast (right minus left)	Accuracy delta: mean ± SD (pp)	Macro-F1 delta (pp)
representation_vs_logmel64; D_sr32_fmax16_logmel64 → D_sr32_fmax16_logmel32	+6.030 ± 1.256	+6.268
representation_vs_logmel64; D_sr32_fmax16_logmel64 → D_sr32_fmax16_mfcc40	-1.508 ± 2.718	-1.034
representation_vs_logmel64; D_sr32_fmax8_logmel64 → D_sr32_fmax8_loglinear64	-0.503 ± 1.569	-0.653
representation_vs_logmel64; D_sr32_fmax8_logmel64 → D_sr32_fmax8_logmel32	+4.439 ± 2.756	+4.768
representation_vs_logmel64; D_sr32_fmax8_logmel64 → D_sr32_fmax8_mfcc40	-1.424 ± 1.596	-0.609
sample_rate_matched_bandwidth; D_sr16_fmax8_loglinear64 → D_sr32_fmax8_loglinear64	-0.838 ± 0.145	-0.995
sample_rate_matched_bandwidth; D_sr16_fmax8_logmel32 → D_sr32_fmax8_logmel32	+0.084 ± 1.635	+0.033
sample_rate_matched_bandwidth; D_sr16_fmax8_logmel64 → D_sr32_fmax8_logmel64	-0.335 ± 0.145	-0.221
sample_rate_matched_bandwidth; D_sr16_fmax8_mfcc40 → D_sr32_fmax8_mfcc40	+0.586 ± 0.384	+0.547
specaugment_vs_clean_cnn; B_cnn_freq → C_specaugment	-0.754 ± 1.005	-0.643

Project evidence index

The paths below refer to the accompanying ESC-50 project. They remain useful provenance references when the PDF is shared, but the project files are not embedded in this PDF. Public web references in the report are clickable.

[P01] `docs/bilingual_20261028/guide_en.md`

[P02] `results/colab_matrix_20261028/analysis/groups.csv`

[P03] `tools/plot_colab_matrix.py`

[P04] `results/colab_matrix_20261028/analysis/paired_contrasts.csv`

[P05] `results/colab_matrix_20261028/analysis/stress.csv`

[P06] `results/colab_matrix_20261028/figures/learning_curve_data.csv`

[P07] `results/colab_matrix_20261028/analysis/runs.csv`

[P08] `results/colab_matrix_20261028/analysis/audit.json`

[P09] `results/colab_matrix_20261028/runtime_verification/waveform_replay.json`

[P10] `results/colab_matrix_20261028/runtime_verification/resume_probe.json`

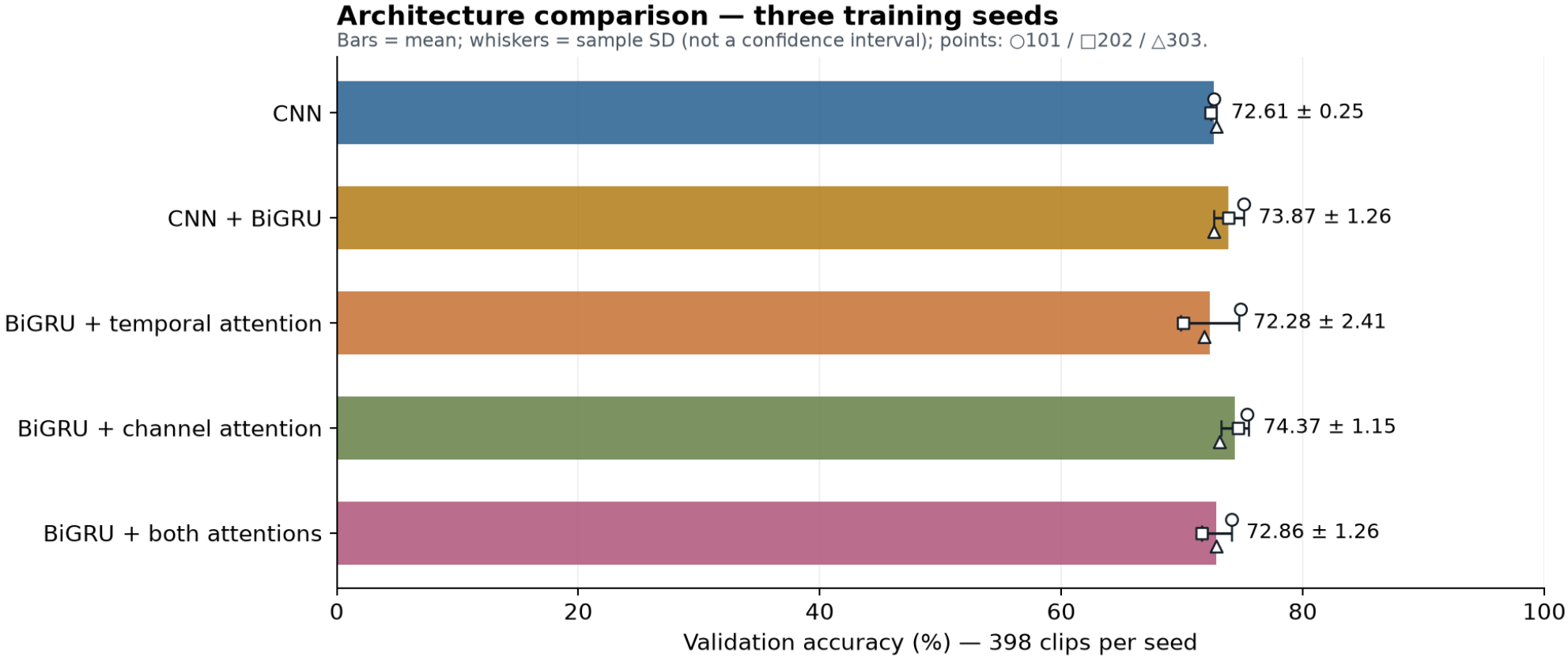
[P11] `results/colab_matrix_20261028/runtime_verification/warm_latency.json`

[P12] `docs/literature_deep_read.md`

[P13] `docs/bilingual_20261028/review_coverage.md`

[P14] `results/colab_matrix_20261028/runtime_verification/measurement_environment.json`

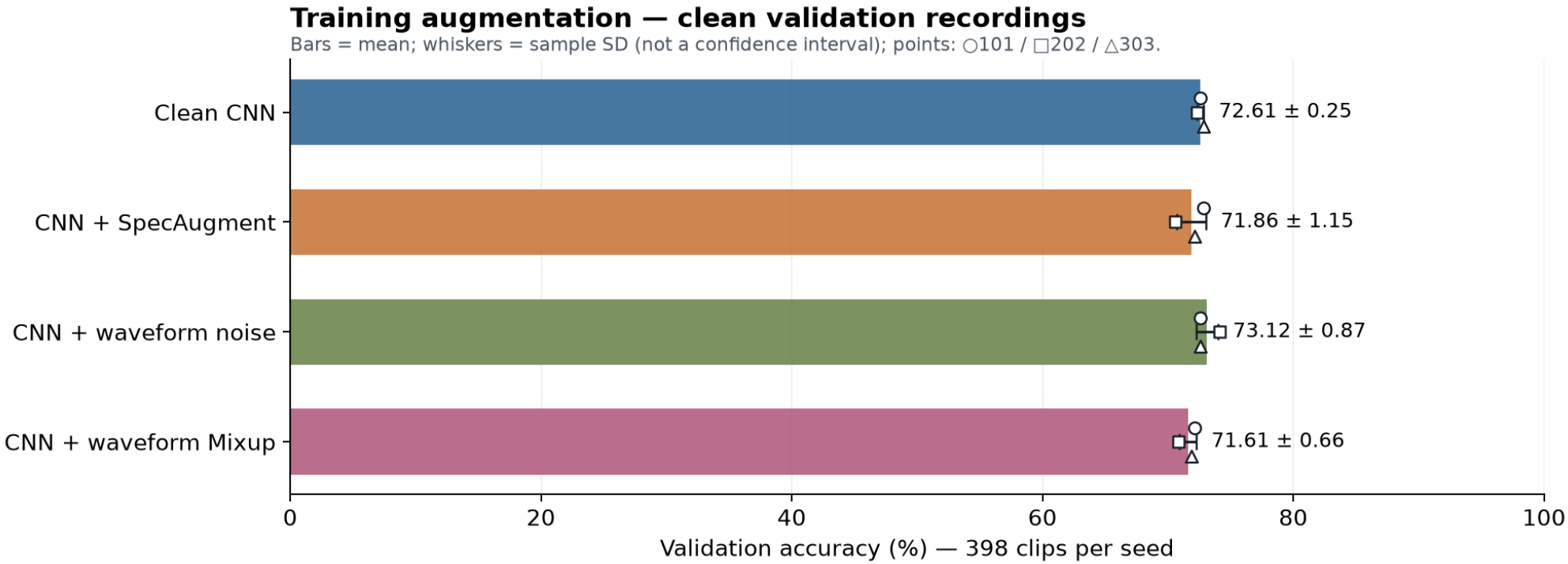
Figure 1. Three-seed development validation; SD is not a confidence interval.



Three-seed development validation; SD is not a confidence interval.

Source: `results/colab_matrix_20261028/figures/architecture.png`. Original saved figure; no experiment was rerun for this PDF.

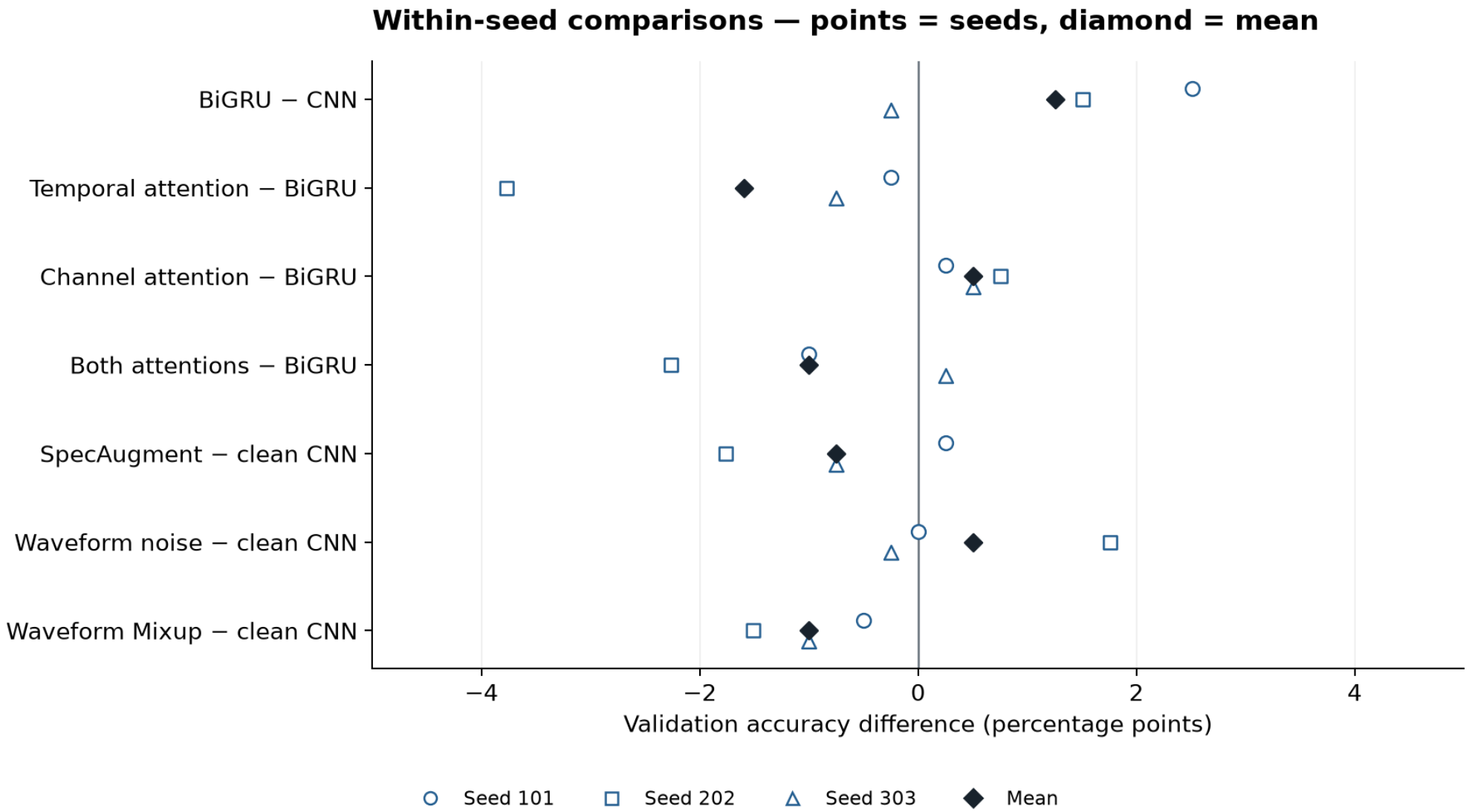
Figure 2. Augmentation affects training only; validation recordings stay clean.



Augmentation affects training only; validation recordings stay clean.

Source: `results/colab_matrix_20261028/figures/augmentation.png`. Original saved figure; no experiment was rerun for this PDF.

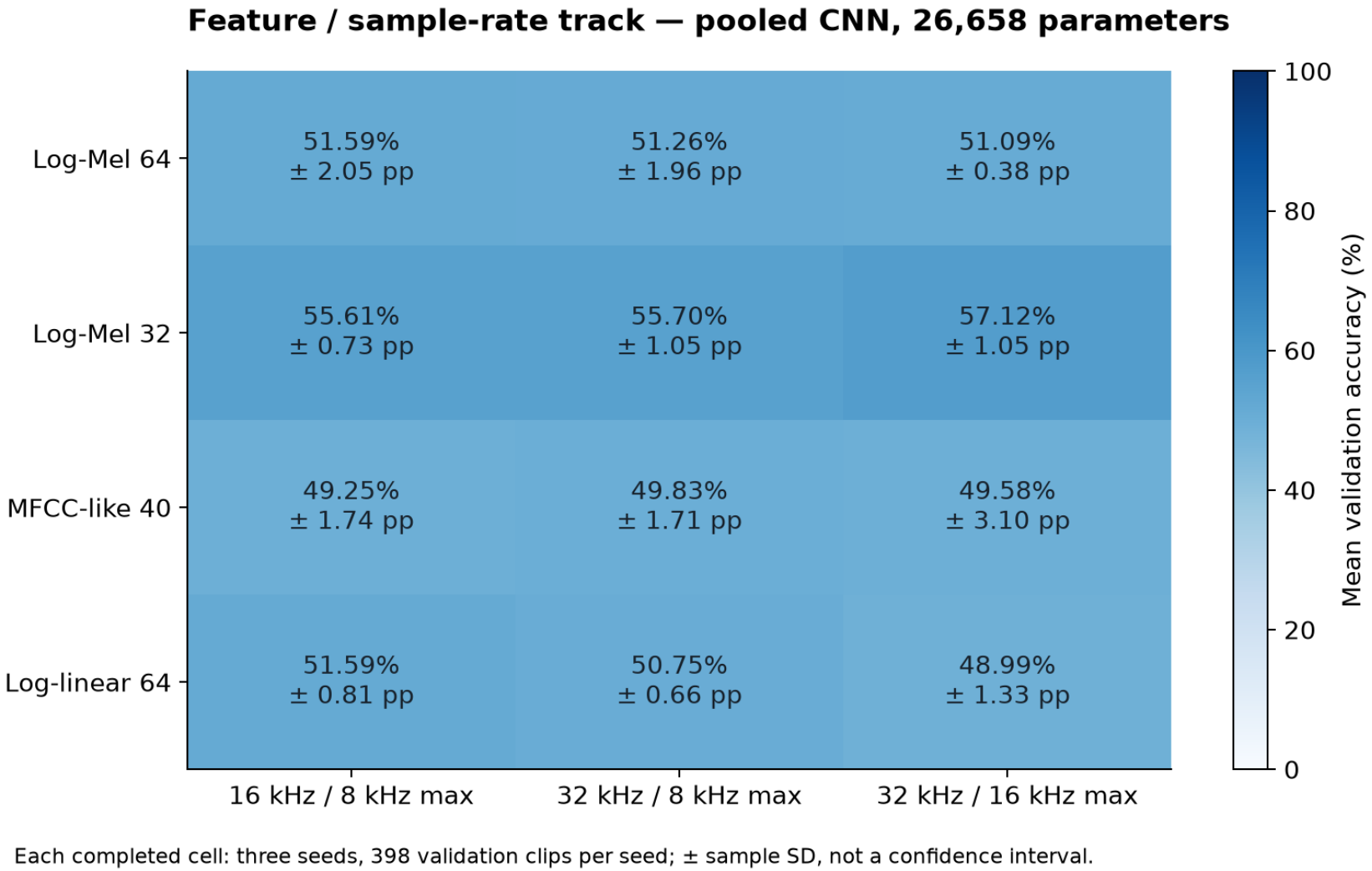
Figure 3. Within-seed deltas; these are not significance tests.



Within-seed deltas; these are not significance tests.

Source: `results/colab_matrix_20261028/figures/paired_effects.png`. Original saved figure; no experiment was rerun for this PDF.

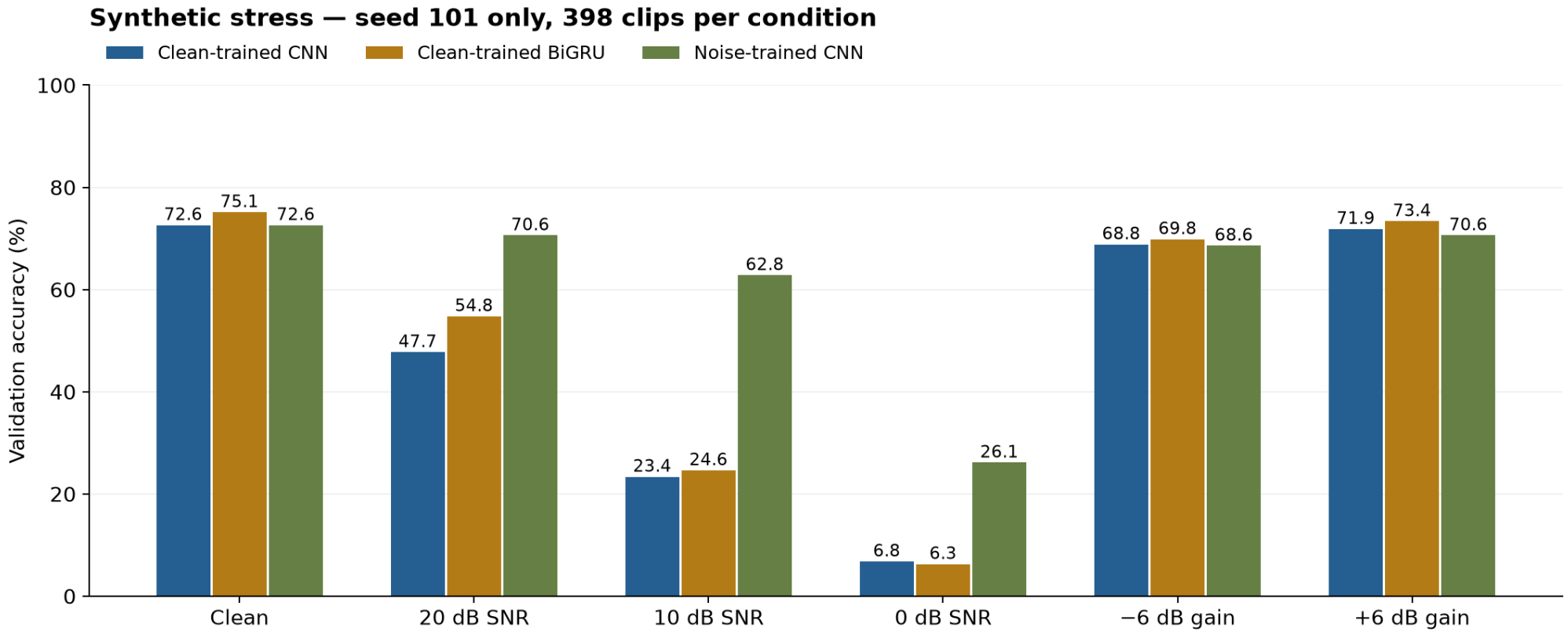
Figure 4. Track D uses the 26,658-parameter pooled CNN; interpret separately from the main architecture track.



Track D uses the 26,658-parameter pooled CNN; interpret separately from the main architecture track.

Source: `results/colab_matrix_20261028/figures/feature_sampling.png`. Original saved figure; no experiment was rerun for this PDF.

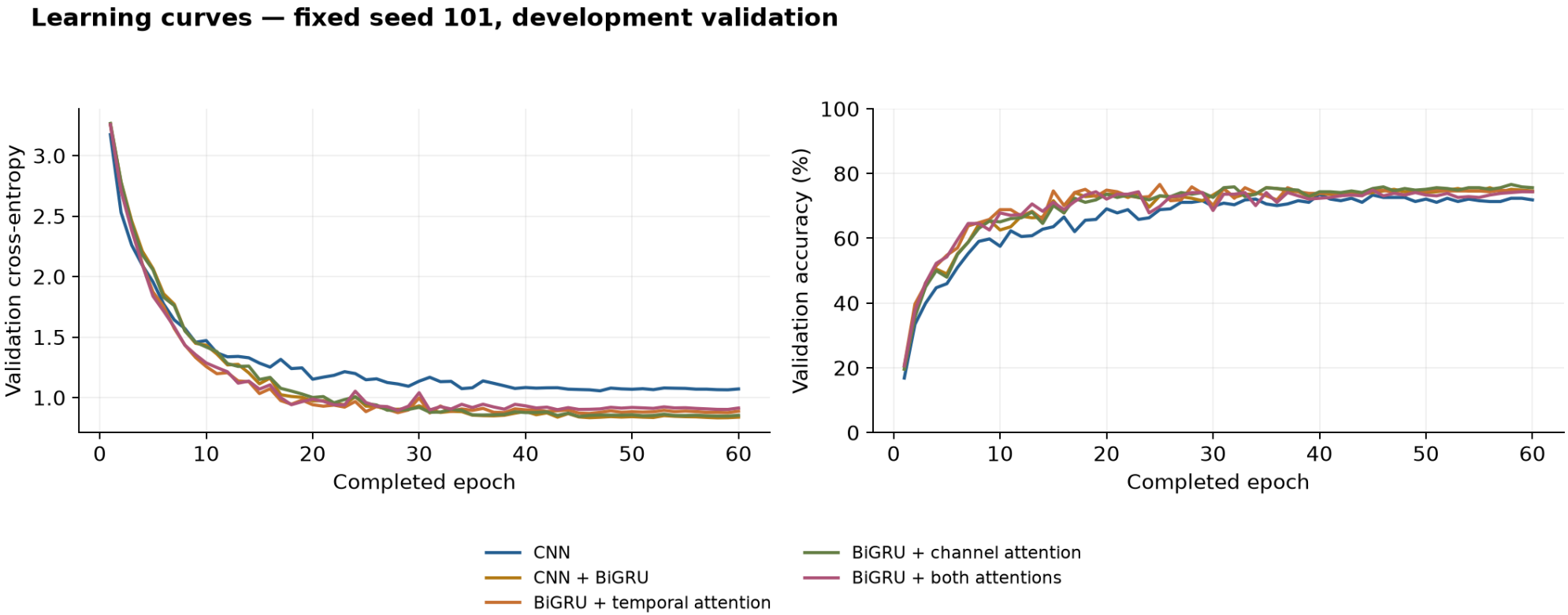
Figure 5. 398 clips per cell; one selected seed101 checkpoint per model.



398 clips per cell; one selected seed101 checkpoint per model.

Source: `results/colab_matrix_20261028/figures/robustness.png`. Original saved figure; no experiment was rerun for this PDF.

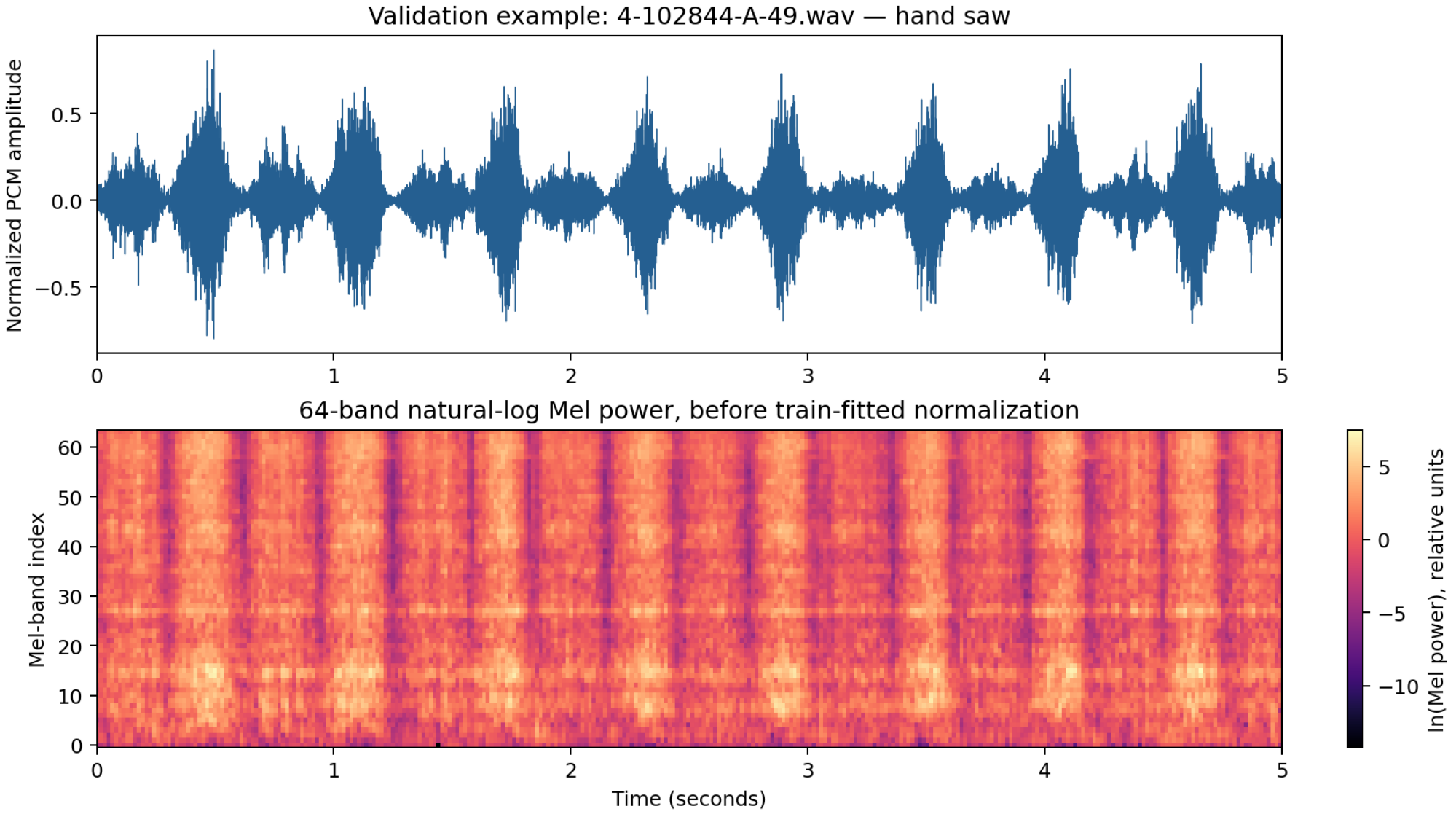
Figure 6. Fixed-seed101 validation curves; checkpoints use minimum cross-entropy.



Fixed-seed101 validation curves; checkpoints use minimum cross-entropy.

Source: `results/colab_matrix_20261028/figures/learning_curves.png`. Original saved figure; no experiment was rerun for this PDF.

Figure 7. A real validation waveform and Log-Mel representation



Five-second hand-saw validation example; visualization only, not an extra training or test result.

Source: `results/colab_matrix_20261028/runtime_verification/frontend_example.png`. Original saved figure; no experiment was rerun for this PDF.